

12.2 Revision creation

The system creates a new draft version by copying the prior version's controlled data.

The new version must:

Receive the next version number.

Reference the prior version.

Include a revision reason.

Remain separate from the original.

Require approval before becoming current.

Identify affected documents and signatures.

12.3 Effect on documents

If the revision affects a signed or approved document:

The original document remains preserved.

The original signature audit record remains attached.

The revised document receives a new version.

The revised document enters a new signature sequence where required.

The new document is marked current or superseding after approval.

13. Task Data Model

workflow_task

Field	Type	Requirement
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id	BIGINT	Primary key
----	--------	-------------

project_id	BIGINT	Required foreign key
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msr_version_id	BIGINT	Optional foreign key
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task_type	VARCHAR(80)	Required
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task_status	VARCHAR(30)	Open, assigned, completed, cancelled, blocked, expired
-------------	-------------	--

assigned_user_id	BIGINT	Optional foreign key
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assigned_role_code	VARCHAR(60)	Optional
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created_at	TIMESTAMPTZ	Required
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created_by	BIGINT	Required foreign key
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assigned_at	TIMESTAMPTZ	Optional
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due_at	TIMESTAMPTZ	Optional
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completed_at	TIMESTAMPTZ	Optional
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completed_by	BIGINT	Optional foreign key
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blocked_reason	TEXT	Optional
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completion_notes	TEXT	Optional
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priority	VARCHAR(20)	Normal, high, urgent
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related_document_id	BIGINT	Optional foreign key
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related_version_id	BIGINT	Optional foreign key
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audit_event_id	BIGINT	Optional foreign key
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Initial task types

text

sales_record_review

sales_record_correction

contract_customer_signature

contract_gecc_signature

invoice_gecc_signature

invoice_customer_signature

installation_scheduling

installation_assignment

installation_completion
scope_change_review
scope_change_revision
certificate_customer_signature
certificate_gecc_signature
payment_entry
payment_verification
adjustment_approval
project_completion_review
exception_review

Tasks are not a substitute for state validation. Both task status and state-transition guards must be satisfied.

14. Project Completion Rules

A project may enter completed only when all conditions are true:

Scope of work is complete.
Contract has all required signatures.
Certificate has all required signatures.
Invoice has all required signatures.
Invoice is fully paid after approved adjustments.
Payment verification is complete.
No unresolved scope change exists.
No required task remains open or blocked.
The current Master Sales Record version is approved.
The project is not cancelled.
Required audit events were created.
The system must reject completion if any condition is false.

15. Exceptions and Blocking Conditions

The system must identify and block:

Missing required approval
Missing required signature
Invalid signing order
Expired signature request
Installation before the waiting period
Missing installation assignment
Missing equipment serial numbers where required
Proposed scope change not reviewed
Revised scope not approved
Unverified payment
Unapproved adjustment
Unpaid remaining balance
Superseded Master Sales Record
Cancelled project
Missing certificate signature
Missing required completion task
A blocking condition must display:

Condition
Affected record
Required action

Assigned role or user
Date detected
Due date where applicable
16. Audit Events
Sprint 06 must create audit events for:

State transition
Transition rejection
Task creation
Task assignment
Task reassignment
Task completion
Task cancellation
Task blocking
Contract-signature dependency validation
Invoice-signature dependency validation
Earliest-installation-date calculation
Early-installation attempt
Installation scheduling
Installation rescheduling
Installation start
Installation completion
Scope-change proposal
Scope-change approval
Scope-change rejection
Scope-change work-block
Authorized reopening
New Master Sales Record version
Revised-document requirement
Exception request
Exception approval
Exception rejection
Project-completion validation
Project-completion rejection
Project completion
Each event must identify:

User
Role
Session
Project
Current state
Previous state
Requested state
Task
Master Sales Record version
Related document or payment where applicable
Outcome
Reason
Timestamp
Audit-chain references
17. Sprint 06 Decisions
D-06-001 — Controlled state machine

Project lifecycle transitions are controlled by a defined state machine.

Users cannot bypass transition guards through the user interface or ordinary application operations.

D-06-002 — Three-day waiting period

The earliest installation date is calculated from the later of the customer contract-signature timestamp and customer invoice-signature timestamp, plus three calendar days.

D-06-003 — Waiting-period timezone

Waiting-period calculations use America/New_York.

D-06-004 — Core approval authority

The Master Sales Record approval authority remains:

Sales Manager, or

Comptroller

One authorized approval is sufficient.

D-06-005 — Installation task creation

An Installation Manager scheduling task is created only after all required contract and invoice signatures are complete.

D-06-006 — Installation blocking

Installation cannot be scheduled, started, or completed before the permitted date unless a separately authorized exception is approved.

D-06-007 — Scope-change blocking

Work on changed scope is blocked while the scope change is pending review or while revised approvals and signatures are incomplete.

D-06-008 — Scope-change versioning

An approved scope change creates a new Master Sales Record version and revised document workflow.

D-06-009 — Completion gate

Project completion requires scope completion, required signatures, verified payment, full payment after approved adjustments, and no unresolved blocking task.

D-06-010 — Historical preservation

State transitions, tasks, exceptions, approvals, and rejected transitions remain permanently auditable.

18. Sprint 06 Acceptance Criteria

Sprint 06 is accepted when:

A project cannot skip required lifecycle states.

A submitted Master Sales Record requires authorized review.

An unauthorized user cannot approve a Master Sales Record.

An approved Master Sales Record becomes locked.

An approved record cannot be silently overwritten.

Authorized reopening creates a new version.

Original Master Sales Record versions remain preserved.

Contract customer signing is required before GECC contract signing.
Contract completion is required before invoice signature workflow.
GECC invoice signing is required before customer invoice signing.
Installation scheduling cannot occur before all required contract and invoice signatures.
The three-day installation rule is calculated correctly.
Installation before the permitted date is rejected.
Early-installation attempts are audited.
Authorized exceptions require a reason and approval.
Installation tasks are created for the Installation Manager.
Installation assignment and scheduling are tracked.
Technicians can update authorized completion fields.
Technicians cannot change retail price, approved costs, or contract terms.
A proposed scope change enters a blocking state.
Changed-scope work cannot continue before approval.
An approved scope change creates a new Master Sales Record version.
Revised documents are required when affected.
Original documents and signatures remain preserved.
Payment verification is required before completion.
Approved adjustments are required before adjusted payment completion.
A project cannot be completed with an unresolved scope change.
A project cannot be completed with an unverified payment.
A project cannot be completed with missing required signatures.
All transitions, tasks, exceptions, and failures are audited.
Workflow tasks expose the next required action and assigned role or user.
The implementation is ready for Sprint 07 document generation.

19. Sprint 06 Test Cases

State-transition tests

Draft transitions to review.
Review transitions to approved.
Review can return to correction.
Unapproved records cannot generate final documents.
Approved records cannot be edited directly.
Cancelled projects cannot resume ordinary processing.
Invalid transitions are rejected.
Rejected transitions are audited.

Signature-order tests

Customer can sign the contract first.
GECC contract signature is blocked until customer signature exists.
GECC invoice signature is blocked until contract is fully signed.
Customer invoice signature is blocked until GECC invoice signature exists.
Certificate customer signature is required before GECC certificate signature.
Comptroller substitute certificate signing requires a reason.

Waiting-period tests

Contract customer signature timestamp is recorded.
Invoice customer signature timestamp is recorded.
Later timestamp is selected.
Three calendar days are added in Eastern time.
Installation before the computed date is rejected.
Installation on the permitted date succeeds.
Exception approval permits only the specifically authorized exception.
Original timestamps remain unchanged.

Scope-change tests

Technician can propose a scope change.

Scope-change proposal blocks changed-scope work.
Sales Manager can approve the change.
Comptroller can approve the change.
Rejected change returns to the original approved scope.
Approved change creates a new Master Sales Record version.
Original version remains unchanged.
Revised approval is required.
Revised document signatures are required where applicable.

Completion tests

Completion is rejected without completed scope.
Completion is rejected without required signatures.
Completion is rejected without verified payment.
Completion is rejected with an outstanding adjusted balance.
Completion is rejected with an open scope change.
Completion succeeds only when every required condition is satisfied.
Completion makes the project eligible for later commission processing.

Task tests

Required task is created on the correct transition.
Task is assigned to the correct role.
Task cannot be completed by an unauthorized user.
Task completion updates the workflow state only when all guards pass.
Blocked tasks display a reason.
Overdue tasks are visible to authorized dashboard users.

20. Sprint 06 Open Questions

These implementation questions do not change approved requirements:

What exact due-date rules apply to each task type?

Are waiting periods counted as calendar days or business days? The current approved rule is three days; the precise operational interpretation requires confirmation.

What exception roles may approve early installation?

What circumstances qualify as Installation Manager unavailability?

Which document changes always require a new signature sequence?

Which scope changes require revised pricing?

Which scope changes require revised costs?

What notification channels will be used for task assignments?

What escalation rules apply to overdue tasks?

What exact state is used while an expired signature request awaits authorized action?

What cancellation rules apply after documents have been signed?

What state should be used for a project that is complete in scope but unpaid?

21. Sprint 07 Handoff

Sprint 07 will implement document generation and artwork.

Sprint 07 must use:

The approved Master Sales Record version

The current project state

The current document-generation task

Document-version references

Required approval state

Customer and project consistency checks

State-specific project location

Scope and equipment data

Retail price

Payment option
Cost data where permitted
Current revision and supersession information
Sprint 07 must not generate final controlled documents when:

The Master Sales Record is not approved.
Required fields are missing.
The record is superseded.
A required workflow approval is missing.
Customer, project, or project-code consistency fails.
Required artwork is missing.
A document must be regenerated from a newer current version.

22. Historical Record

Supersedes

None.

Preserves

GECC-00-project-baseline.md

GECC-01-data-model.md

GECC-02-auth-and-permissions.md

GECC-03-audit-log.md

GECC-04-customers-and-projects.md

GECC-05-master-sales-record.md

Sprint 00 decisions

Sprint 01 decisions

Sprint 02 decisions

Sprint 03 decisions

Sprint 04 decisions

Sprint 05 decisions

Approved lifecycle requirements

Approved waiting-period requirement

Approved scope-change requirements

Approved versioning and audit requirements

Changes introduced by this bundle

Defines the controlled project state machine.

Defines allowed lifecycle transitions.

Defines transition guards.

Defines task creation and assignment.

Defines contract and invoice dependencies.

Defines the three-day installation rule.

Defines installation scheduling and completion workflow.

Defines scope-change blocking and revision workflow.

Defines authorized reopening.

Defines project-completion gates.

Defines Sprint 06 audit events.

Defines Sprint 07 document-generation handoff.

===== END FILE: GECC-06-Workflow-State-Machine.md =====

=== BEGIN FILE: GECC-07-Document-Generation.md ===

GECC-07-document-generation.md

GECC Sales Back Office System — Sprint 07 Document Generation and Artwork

Bundle Metadata

- Bundle name: GECC-07-document-generation.md
- Project: GECC Sales Back Office System
- Company: Go Ecco Climate Control
- Company abbreviation: GECC
- Bundle type: Sprint 07 design and implementation bundle
- Status: Approved design; ready for implementation
- Created date: 2026-09-20
- Previous bundle: GECC-06-workflow-state-machine.md
- Next planned bundle: GECC-08-electronic-signatures.md
- Database: PostgreSQL
- Primary-key strategy: Numeric BIGINT identity keys
- Business timezone: America/New_York
- System of record: PostgreSQL relational database
- Audit model: Append-only, hash-chained, permanently retained

1. Sprint Objective

Implement controlled generation of:

- Invoices
- Contracts
- Certificates

Documents must be generated from an approved Master Sales Record and stored as immutable document versions in the customer/project file.

The system must support:

- Existing GECC PDF templates
- Automatically generated invoices
- GECC logo placement
- State-specific license artwork
- Artwork versioning
- Document preview
- PDF generation
- Saving
- Printing
- Emailing
- Signature-system handoff
- Document-history preservation

Electronic signature execution is implemented in Sprint 08.

2. Sprint 07 Deliverables

Sprint 07 must produce:

1. Invoice-generation process

2. Contract-template population process
3. Certificate-template population process
4. PDF-template registration
5. Template versioning
6. Document-version model
7. GECC logo management
8. State-specific license-artwork management
9. Base64 artwork validation
10. Effective-dated artwork activation
11. Artwork preview
12. Automatic state-artwork selection
13. Authorized artwork override
14. Document-generation validation
15. PDF storage and file references
16. Document preview
17. Print support
18. Email handoff
19. Electronic-signature handoff interface
20. Document-generation audit events
21. Document consistency tests
22. Sprint 08 signature-system handoff

3. Requirements Carried Forward

The implementation must preserve these approved requirements:

- Documents are generated from the approved Master Sales Record.
- The invoice number equals the project identification code.
- Contract and certificate forms are existing GECC PDF templates.
- Invoices are generated automatically.
- Contracts and certificates are populated automatically.
- Documents can be saved.
- Documents can be printed.
- Documents can be emailed.
- Documents can be sent for electronic signature.
- Generated documents are saved to the customer/project file.
- Original documents remain permanently preserved.
- Revised documents receive new versions.
- Generated documents must not be independently edited in a way that changes underlying project data.
- State-specific license artwork is selected from the project location's state.
- Artwork overrides require alternate-artwork selection, approval reason, Sales Manager or Comptroller approval, and an audit event.
- Document generation must stop when required artwork is missing.
- Document generation must stop when required approvals or data are missing.
- Each document retains its template and artwork version references.
- Document versions remain linked to the associated Master Sales Record version.

4. Document Types

4.1 Invoice

The invoice must include or reference:

- Invoice number
- Project identification code
- Customer name
- Customer address where applicable
- Project location
- Scope of work
- Retail price
- Payment option
- Adjustment amount where applicable
- Adjusted amount due where applicable
- GECC signer fields
- Customer signer fields
- GECC logo
- State-specific license artwork where required
- Master Sales Record version
- Document version
- Template version
- Artwork version

The invoice number must equal the unique project identification code.

4.2 Contract

The contract uses an existing GECC PDF template.

The populated contract must include or reference:

- Customer information
- Project identification code
- Project location
- Scope of work
- Equipment
- Applicable pricing and contract terms
- Required signature fields
- GECC logo
- State-specific license artwork where required
- Master Sales Record version
- Template version
- Document version
- Artwork version

4.3 Certificate

The certificate uses an existing GECC PDF template.

The populated certificate must include or reference:

- Customer information

- Project identification code
- Project location
- Approved scope of work
- Equipment
- Serial numbers
- Installation completion information
- Customer signature fields
- Installation Manager signature fields
- Comptroller substitute-signature fields where applicable
- Substitute-signature reason where applicable
- GECC logo
- State-specific license artwork where required
- Master Sales Record version
- Template version
- Document version
- Artwork version

5. Document Data Model

5.1 `document`

Field	Type	Requirement
---	---	---
`id`	BIGINT	Primary key
`project_id`	BIGINT	Required foreign key
`document_type`	VARCHAR(30)	Invoice, contract, certificate
`current_version_id`	BIGINT	Optional foreign key
`document_status`	VARCHAR(40)	Required
`required_for_workflow`	BOOLEAN	Required
`created_at`	TIMESTAMPTZ	Required
`created_by`	BIGINT	Required foreign key
`updated_at`	TIMESTAMPTZ	Required
`updated_by`	BIGINT	Required foreign key
`is_active`	BOOLEAN	Required
`retired_at`	TIMESTAMPTZ	Optional
`retired_by`	BIGINT	Optional foreign key
`retirement_reason`	TEXT	Optional

Constraint:

```text

UNIQUE(project\\_id, document\\_type)

A project has one logical invoice, one logical contract, and one logical certificate, each of which may have multiple versions.

### 5.2 document\_version

| Field          | Type   | Requirement          |
|----------------|--------|----------------------|
| id             | BIGINT | Primary key          |
| document_id    | BIGINT | Required foreign key |
| project_id     | BIGINT | Required foreign key |
| msr_version_id | BIGINT | Required foreign key |

|                                |              |                                                  |
|--------------------------------|--------------|--------------------------------------------------|
| version_number                 | INTEGER      | Required                                         |
| document_status                | VARCHAR(50)  | Draft, generated, current, superseded, cancelled |
| template_version_id            | BIGINT       | Required foreign key                             |
| artwork_selection_id           | BIGINT       | Required foreign key where artwork is required   |
| file_location                  | TEXT         | Required                                         |
| file_name                      | VARCHAR(255) | Required                                         |
| content_type                   | VARCHAR(100) | Required                                         |
| content_hash                   | VARCHAR(128) | Required                                         |
| file_size_bytes                | BIGINT       | Required                                         |
| generated_at                   | TIMESTAMPTZ  | Required                                         |
| generated_by                   | BIGINT       | Required foreign key                             |
| is_current                     | BOOLEAN      | Required                                         |
| is_signed                      | BOOLEAN      | Required                                         |
| supersedes_document_version_id | BIGINT       | Optional foreign key                             |
| signature_request_id           | BIGINT       | Optional foreign key                             |
| signature_audit_certificate_id | BIGINT       | Optional foreign key                             |
| Standard audit columns         | —            | Required                                         |

Constraints:

text

UNIQUE(document\\_id, version\\_number)  
Only one document version may be current for a logical document.

### 5.3 document\_field\_value

| Field                 | Type         | Requirement          |
|-----------------------|--------------|----------------------|
| id                    | BIGINT       | Primary key          |
| document_version_id   | BIGINT       | Required foreign key |
| field_name            | VARCHAR(150) | Required             |
| source_entity_type    | VARCHAR(100) | Required             |
| source_entity_id      | BIGINT       | Required             |
| source_field_name     | VARCHAR(150) | Required             |
| rendered_value_hash   | VARCHAR(128) | Required             |
| rendered_value_masked | TEXT         | Optional             |
| created_at            | TIMESTAMPTZ  | Required             |

This table provides traceability from rendered document fields to source records without unnecessarily storing sensitive values.

## 6. Template Management

### 6.1 document\_template

| Field                       | Type         | Requirement                    |
|-----------------------------|--------------|--------------------------------|
| id                          | BIGINT       | Primary key                    |
| template_type               | VARCHAR(30)  | Invoice, contract, certificate |
| template_name               | VARCHAR(200) | Required                       |
| description                 | TEXT         | Optional                       |
| is_active                   | BOOLEAN      | Required                       |
| Standard retirement columns | —            | Required                       |

### 6.2 document\_template\_version

| Field                | Type    | Requirement          |
|----------------------|---------|----------------------|
| id                   | BIGINT  | Primary key          |
| document_template_id | BIGINT  | Required foreign key |
| version_number       | INTEGER | Required             |

|                                |              |                      |                        |
|--------------------------------|--------------|----------------------|------------------------|
| file_location                  | TEXT         | Required             |                        |
| content_type                   | VARCHAR(100) | Required             |                        |
| content_hash                   | VARCHAR(128) | Required             |                        |
| field_map_definition           | JSONB        | Required             |                        |
| signature_placement_definition | JSONB        | Optional             |                        |
| effective_from                 | DATE         | Required             |                        |
| effective_to                   | DATE         | Optional             |                        |
| template_status                | VARCHAR(30)  |                      | Draft, active, retired |
| uploaded_by                    | BIGINT       | Required foreign key |                        |
| uploaded_at                    | TIMESTAMPTZ  | Required             |                        |
| retired_by                     | BIGINT       | Optional foreign key |                        |
| retirement_reason              | TEXT         | Optional             |                        |

Constraints:

Template versions cannot overlap for the same template type and effective scope.  
A template version used by a document remains preserved.  
Retiring a template does not invalidate historical documents.

### 6.3 Template registration

The system must support:

- Uploading an existing GECC contract PDF
- Uploading an existing GECC certificate PDF
- Registering an invoice template or invoice-rendering definition
- Defining field placements
- Defining artwork placements
- Defining signature placements
- Previewing a template
- Activating a template version
- Retiring a template version
- Preserving the template version used by each document

The actual contract and certificate files remain an open implementation input until supplied by GECC.

## 7. Invoice Generation

### 7.1 Invoice creation

An invoice may be generated only when:

- The Master Sales Record is approved.
- The selected Master Sales Record version is current and approved.
- Required customer data exists.
- Required project data exists.
- Project-code consistency passes.
- Retail price is present.
- Payment option is present.
- Required artwork exists.
- Required template exists.
- No blocking workflow condition exists.

### 7.2 Invoice number

The invoice number must equal:

text

project.project\\_code

The system must prevent:

Duplicate invoice numbers

Invoice-number changes after generation

Invoice generation from a mismatched project code

Invoice generation from a superseded Master Sales Record version

### 7.3 Invoice amount

The initial invoice amount is sourced from the approved retail price.

Approved adjustments may later produce:

Original invoice amount

Adjustment amount

Adjusted amount due

The original invoice amount remains preserved.

### 7.4 Invoice versioning

A new invoice version is required when:

The Master Sales Record is revised.

Retail price changes.

Payment terms change.

Required invoice data changes.

Approved adjustments require a revised invoice.

A document-generation correction is authorized.

The original invoice version remains preserved.

## 8. Contract Generation

### 8.1 Contract prerequisites

The contract may be generated only when:

The Master Sales Record is approved.

The required contract template is active.

Required customer fields exist.

Required project fields exist.

Scope data is complete.

Contract field mapping is valid.

State-specific artwork is available or an approved override exists.

Required signature fields are mapped.

### 8.2 Contract population

The system must populate the approved template from controlled data.

The system must not require the Sales Associate to re-enter data already stored in the Master Sales Record.

### 8.3 Contract review

After generation:

The Sales Associate may review the generated contract.

The Sales Associate may report an error.

The Sales Associate may not directly alter the PDF to change source data.

Corrections must be made to the source record or authorized template configuration.

Regeneration creates a new document version where appropriate.

## 9. Certificate Generation

### 9.1 Certificate prerequisites

The certificate may be generated when:

The approved Master Sales Record exists.

Required project and customer data exists.

Approved scope exists.

Equipment and serial-number fields required for the certificate are present.

The certificate template is active.

Required artwork exists or an approved override exists.

Required signature fields are mapped.

### 9.2 Certificate updates

The certificate may use completion information recorded later by the Technician, including:

Actual equipment serial numbers

Actual materials used

Installation completion date

Completion notes

Approved scope version

A certificate generated before completion may be regenerated after required completion data becomes available.

### 9.3 Certificate versioning

A certificate regenerated after a scope or source-data change receives a new version.

The original certificate remains preserved.

## 10. Artwork Management

### 10.1 artwork\_asset

| Field | Type | Requirement |
|-------|------|-------------|
|-------|------|-------------|

|                             |              |                                    |
|-----------------------------|--------------|------------------------------------|
| id                          | BIGINT       | Primary key                        |
| artwork_type                | VARCHAR(40)  | GECC logo, state license           |
| name                        | VARCHAR(200) | Required                           |
| state_code                  | CHAR(2)      | Required for state license artwork |
| file_location               | TEXT         | Required                           |
| base64_content_hash         | VARCHAR(128) | Required                           |
| mime_type                   | VARCHAR(100) | Required                           |
| file_extension              | VARCHAR(20)  | Required                           |
| width_pixels                | INTEGER      | Required where applicable          |
| height_pixels               | INTEGER      | Required where applicable          |
| artwork_status              | VARCHAR(30)  | Draft, active, retired             |
| uploaded_by                 | BIGINT       | Required foreign key               |
| uploaded_at                 | TIMESTAMPTZ  | Required                           |
| Standard retirement columns | —            | Required                           |

### 10.2 artwork\_version

| Field | Type | Requirement |
|-------|------|-------------|
|-------|------|-------------|

|                  |         |                      |
|------------------|---------|----------------------|
| id               | BIGINT  | Primary key          |
| artwork_asset_id | BIGINT  | Required foreign key |
| version_number   | INTEGER | Required             |
| effective_from   | DATE    | Required             |
| effective_to     | DATE    | Optional             |
| is_active        | BOOLEAN | Required             |

approval\_status VARCHAR(30) Required  
approved\_by BIGINT Optional foreign key  
approved\_at TIMESTAMPTZ Optional  
retired\_by BIGINT Optional foreign key  
retirement\_reason TEXT Optional  
Standard audit columns — Required

### 10.3 Artwork requirements

The system must support:

Upload of Base64 artwork  
Approved image-format validation  
GECC logo artwork  
State-specific license artwork  
State association  
Effective date  
Activation  
Retirement  
Preview before activation  
Historical preservation  
Document-level artwork references

### 10.4 Base64 validation

The system must validate:

Valid Base64 encoding  
Permitted MIME type  
Permitted image format  
Decodable image content  
Non-empty content  
Maximum file size  
Image dimensions  
Content hash  
Malware or unsafe-file screening where supported  
The approved format list remains a configuration item.

### 10.5 Automatic state selection

For a project in state XX, the system selects the active artwork version where:

text

artwork\\_type = state\\_license  
state\\_code = XX  
effective\\_from <= document\\_date  
effective\\_to is null or effective\\_to >= document\\_date  
approval\\_status = approved  
is\\_active = true  
The selected artwork version is stored on the document version.

### 10.6 Artwork override

An authorized user may override automatic selection only when:

Alternate artwork is selected.  
An approval reason is entered.

Sales Manager or Comptroller approval is recorded.  
The override is linked to the document-generation request.  
An audit event is created.  
Document generation must stop if the override is incomplete.

#### 10.7 Missing artwork

Document generation must stop if:

Required state artwork is missing.  
No active artwork applies to the project state.  
Artwork is not approved.  
Artwork is expired.  
Artwork content fails validation.  
An override lacks approval or reason.  
The artwork version cannot be preserved.

#### 11. Artwork Selection

document\_artwork\_selection

| Field | Type | Requirement |
|-------|------|-------------|
|-------|------|-------------|

|    |        |             |
|----|--------|-------------|
| id | BIGINT | Primary key |
|----|--------|-------------|

|                     |        |                      |
|---------------------|--------|----------------------|
| document_version_id | BIGINT | Required foreign key |
|---------------------|--------|----------------------|

|                    |         |          |
|--------------------|---------|----------|
| project_state_code | CHAR(2) | Required |
|--------------------|---------|----------|

|                              |        |                      |
|------------------------------|--------|----------------------|
| automatic_artwork_version_id | BIGINT | Optional foreign key |
|------------------------------|--------|----------------------|

|                             |        |                      |
|-----------------------------|--------|----------------------|
| selected_artwork_version_id | BIGINT | Required foreign key |
|-----------------------------|--------|----------------------|

|                  |             |                              |
|------------------|-------------|------------------------------|
| selection_method | VARCHAR(30) | Automatic, approved override |
|------------------|-------------|------------------------------|

|                 |      |                       |
|-----------------|------|-----------------------|
| override_reason | TEXT | Required for override |
|-----------------|------|-----------------------|

|                      |        |                       |
|----------------------|--------|-----------------------|
| override_approved_by | BIGINT | Required for override |
|----------------------|--------|-----------------------|

|                      |             |                       |
|----------------------|-------------|-----------------------|
| override_approved_at | TIMESTAMPTZ | Required for override |
|----------------------|-------------|-----------------------|

|             |             |          |
|-------------|-------------|----------|
| selected_at | TIMESTAMPTZ | Required |
|-------------|-------------|----------|

|             |        |                      |
|-------------|--------|----------------------|
| selected_by | BIGINT | Required foreign key |
|-------------|--------|----------------------|

Each document retains:

Selected state

Artwork version

Artwork effective date

Uploading or changing user

Date and time used

Selection method

Override reason where applicable

#### 12. Document-Generation Validation

Before generation, the system must validate:

Required fields

Customer name

Customer contact data where required

Project code

Project location

State

Scope

Equipment where required

Retail price where required

Payment option

Required signer fields

Required template

Required artwork

Consistency

Customer matches the project.

Project matches the Master Sales Record.

Project code matches the invoice number where applicable.

Current Master Sales Record version is used.

Scope matches the approved version.

Equipment and serial numbers match the approved or current authorized data.

Pricing matches the approved version.

Document references the correct version.

State artwork matches the project location unless an approved override exists.

Workflow

Required approval exists.

Required task or state permits generation.

The project is not cancelled.

The source version is not superseded.

A revised document is generated when a required source version changes.

Document generation must fail with actionable validation messages.

### 13. Document Rendering

The rendering process must:

Load the current approved source version.

Load the active template version.

Resolve document field mappings.

Resolve the state-specific artwork.

Validate all required fields.

Render the document.

Create a content hash.

Save the PDF to controlled file storage.

Create the document-version record.

Save source-field traceability references.

Mark the document version generated.

Create the audit event.

Return the preview or saved-file reference.

The source Master Sales Record must not be modified by rendering.

### 14. File Storage

Generated documents must be stored in the customer/project file structure.

The file system or object store must preserve:

Project identifier

Document type

Document version

Master Sales Record version

Generation timestamp

Content hash

Current or superseded status

Signature status

File location

Access restrictions

Recommended logical path:

text

/customer/{customer\\_id}/project/{project\\_id}/{document\\_type}/  
Recommended file naming pattern:

text

{project\\_code}-{document\\_type}-v{version\\_number}.pdf  
The exact physical storage technology remains an implementation question.

## 15. Save, Preview, Print, and Email

Authorized forms must support:

Save source data  
Save draft  
Generate document  
Preview document  
Save PDF  
Print document  
Email document  
Send document for electronic signature  
Clear form  
Cancel form

### 15.1 Preview

Preview must:

Use the same data and template as final generation.  
Display selected artwork.  
Display document version information where configured.  
Not create a final current document version unless explicitly saved.  
Record preview activity in the audit log.

### 15.2 Print

Printing must:

Use the generated document version.  
Respect user authorization.  
Record the print event.  
Preserve the printed document's content hash where available.

### 15.3 Email

Emailing must:

Use an authorized document version.  
Record sender, recipient, timestamp, document version, and result.  
Prevent unauthorized recipients where recipient rules apply.  
Preserve the emailed document version.  
Not silently replace a current document with an older version.  
15.4 Electronic-signature handoff  
Sprint 07 creates the document and signature-handoff request data.

Sprint 08 implements:

Signature delivery  
Signer authentication  
Signing order  
Reminders  
Expiration  
Audit certificates  
Signed PDF return

#### 16. Document Version Rules

A document version is immutable after generation.

The system must preserve:

Source Master Sales Record version  
Template version  
Artwork version  
Document version  
File location  
Content hash  
Generated timestamp  
Generating user  
Field-source traceability  
Signature references  
Superseded-document relationship  
A revised document:

Receives a new version number.  
References the version it supersedes.  
Is generated from the new approved source version.  
Does not alter the original document.  
Enters the required signature process where applicable.

#### 17. Audit Events

Sprint 07 must create audit events for:

Template uploaded  
Template previewed  
Template activated  
Template retired  
Artwork uploaded  
Artwork validated  
Artwork previewed  
Artwork activated  
Artwork retired  
Artwork override requested  
Artwork override approved  
Artwork override rejected  
Document-generation request  
Document-generation validation failure  
Document generated  
Document previewed  
Document saved  
Document printed  
Document emailed

Document sent for signature  
Document superseded  
Document cancelled  
Document downloaded  
Unauthorized document-generation attempt  
Missing-artwork failure  
Template-mapping failure  
Source-data mismatch  
Each event must reference:

User  
Role  
Session  
Project  
Customer  
Document  
Document version  
Master Sales Record version  
Template version  
Artwork version where applicable  
Outcome  
Reason

Content hash where applicable

Audit-chain references

18. Sprint 07 Decisions

D-07-001 — Source data

All controlled documents are generated from the approved Master Sales Record version.

D-07-002 — Document immutability

Generated document versions are immutable.

Corrections create new document versions.

D-07-003 — Document traceability

Every document version references:

Master Sales Record version

Template version

Artwork version

Generation user

Generation time

File content hash

D-07-004 — Invoice number

The invoice number equals the project identification code.

D-07-005 — Automatic state artwork

The project location's state determines default license artwork.

D-07-006 — Artwork override

Artwork overrides require alternate artwork, a reason, Sales Manager or Comptroller approval, and an audit event.

D-07-007 — Missing-artwork block

Document generation stops when required artwork is missing, inactive, expired, invalid, or improperly overridden.

#### D-07-008 — Template preservation

A document retains the exact template version used for generation, even after that template is retired.

#### D-07-009 — Signature separation

Sprint 07 generates documents and signature handoff data. Sprint 08 controls the electronic-signature process.

#### D-07-010 — Source-data correction

A PDF may not be manually edited to change underlying project data. Source records must be corrected through the approved workflow and the document regenerated.

### 19. Sprint 07 Acceptance Criteria

Sprint 07 is accepted when:

An approved Master Sales Record can generate an invoice.

An approved Master Sales Record can populate a contract template.

An approved Master Sales Record can populate a certificate template.

The invoice number equals the project identification code.

Documents use the correct customer and project.

Documents use the correct approved Master Sales Record version.

Documents use the correct scope.

Documents use the correct equipment and serial numbers where applicable.

Documents use the correct retail price.

Documents use the correct payment option.

Required fields are validated before generation.

Document generation fails when required data is missing.

Document generation fails when customer, project, or project-code consistency fails.

Document generation fails when required approvals are missing.

Document generation fails when the source version is superseded.

Contract and certificate templates are versioned.

Generated documents are stored in the customer/project file.

Generated documents have content hashes.

Generated document versions are immutable.

Original documents remain preserved after revision.

Revised documents reference their superseded versions.

GECC logo artwork can be applied.

State-specific license artwork is selected automatically.

Artwork is validated before activation.

Artwork versions are effective-dated.

Historical documents retain the artwork version used.

Missing artwork prevents generation.

Unauthorized artwork overrides are rejected.

Authorized artwork overrides require reason and approval.

Preview uses the same source data and template as generation.

Printing is permission-controlled and audited.

Emailing is permission-controlled and audited.

Signature handoff references the exact document version.

All generation and artwork actions are audited.

The document-generation process is ready for Sprint 08 electronic signatures.

## 20. Sprint 07 Test Cases

### Invoice tests

Generate an invoice from an approved Master Sales Record.  
Confirm invoice number equals project code.  
Reject invoice generation from an unapproved record.  
Reject invoice generation from a superseded version.  
Reject invoice generation with missing retail price.  
Confirm payment option appears correctly.  
Confirm the generated PDF is stored and hashed.  
Confirm the invoice version is linked to the source version.

### Contract tests

Upload a contract template.  
Register a template version.  
Populate the contract from approved data.  
Confirm customer data is consistent.  
Confirm project-code data is consistent.  
Confirm scope data is consistent.  
Reject generation with missing template field mapping.  
Retire a template and confirm historical documents remain available.

### Certificate tests

Generate a certificate with approved scope.  
Populate equipment and serial numbers.  
Reject generation when required serial numbers are missing.  
Regenerate after completion information is recorded.  
Preserve the original certificate version.  
Confirm the new certificate references the current source version.

### Artwork tests

Upload valid Base64 artwork.  
Reject invalid Base64 content.  
Reject unsupported image type.  
Reject empty artwork.  
Associate artwork with a state.  
Activate artwork after approval.  
Automatically select artwork based on project state.  
Reject generation when artwork is missing.  
Request an artwork override.  
Reject override without a reason.  
Reject override without Sales Manager or Comptroller approval.  
Confirm the selected artwork version is retained on the document.

### Versioning tests

Generate Version 1 of a document.  
Change the approved source through an authorized revision.  
Generate Version 2.  
Confirm Version 1 remains unchanged.  
Confirm Version 2 references Version 1 as superseded.  
Confirm each version has a different content hash.  
Confirm each version retains its source and template references.

### Access and audit tests

Unauthorized user cannot generate restricted documents.  
Unauthorized user cannot download restricted documents.  
Unauthorized user cannot print restricted documents.  
Unauthorized user cannot approve artwork overrides.  
Document generation is audited.

Preview is audited.

Print is audited.

Email is audited.

Artwork activation is audited.

Validation failures are audited.

## 21. Sprint 07 Open Questions

These implementation questions do not change approved requirements:

What exact contract PDF will GECC provide?

What exact certificate PDF will GECC provide?

What invoice layout and field placement will be approved?

Which PDF-rendering or overlay approach will be used?

Which image formats are approved for logo and license artwork?

What maximum artwork dimensions and file sizes are permitted?

What exact signature field placements will the templates use?

What physical file-storage technology will be used?

What file-retention and backup policy applies to generated documents?

What email service will deliver documents?

What document metadata must appear visibly on each PDF?

What exact recipients may receive each document type?

Which document changes require a new customer signature?

What signature-handoff payload will Sprint 08 require?

What legal or operational wording is required on generated documents?

## 22. Sprint 08 Handoff

Sprint 08 will implement electronic signatures.

Sprint 08 must use:

Document ID

Document version ID

Project ID

Customer ID

Master Sales Record version

Template version

Artwork version

File location

Content hash

Signer roles

Required signing order

Current workflow state

Sprint 08 must preserve:

Original unsigned document

Signature-request version

Delivery history

Reminder history

Expiration history

Completed signed PDF

Signature audit certificate

Relationship between the signed document and source Master Sales Record version

Sprint 08 must not alter the underlying Master Sales Record through signature processing.

## 23. Historical Record

Supersedes  
None.

Preserves

GECC-00-project-baseline.md

GECC-01-data-model.md

GECC-02-auth-and-permissions.md

GECC-03-audit-log.md

GECC-04-customers-and-projects.md

GECC-05-master-sales-record.md

GECC-06-workflow-state-machine.md

Sprint 00 decisions

Sprint 01 decisions

Sprint 02 decisions

Sprint 03 decisions

Sprint 04 decisions

Sprint 05 decisions

Sprint 06 decisions

Approved document, artwork, versioning, workflow, and audit requirements

Changes introduced by this bundle

Defines document-generation architecture.

Defines invoice, contract, and certificate generation.

Defines template and document versioning.

Defines artwork storage, validation, activation, and retirement.

Defines automatic state-artwork selection.

Defines artwork override controls.

Defines document rendering and storage.

Defines print, email, and signature-handoff behavior.

Defines Sprint 07 audit events.

Defines Sprint 08 electronic-signature handoff.

===== END FILE: GECC-07-Document-Generation.md =====

=== BEGIN FILE: GECC-08-Electronic-Signatures.md ===

# GECC-08-electronic-signatures.md

# GECC Sales Back Office System — Sprint 08 Electronic Signatures and Reminders

## Bundle Metadata

- Bundle name: GECC-08-electronic-signatures.md
- Project: GECC Sales Back Office System
- Company: Go Ecco Climate Control
- Company abbreviation: GECC
- Bundle type: Sprint 08 design and implementation bundle
- Status: Approved design; ready for implementation
- Created date: 2026-09-20
- Previous bundle: GECC-07-document-generation.md
- Next planned bundle: GECC-09-payments-and-payables.md
- Database: PostgreSQL
- Primary-key strategy: Numeric BIGINT identity keys
- Business timezone: America/New\_York
- System of record: PostgreSQL relational database
- Audit model: Append-only, hash-chained, permanently retained
- Signature-request expiration: 75 days

- Reminder interval: 7 days

---

## ## 1. Sprint Objective

Implement the built-in electronic-signature system for:

- Contracts
- Invoices
- Certificates

The system must support:

- Email delivery
- Signature links
- Signing order
- Customer authentication
- Signer authentication
- Automatic reminders
- Seven-day reminder intervals
- Seventy-five-day expiration
- Signature audit trails
- Completed-document certificates
- Downloadable signed PDFs
- Signature-status tracking
- Delivery-status tracking
- Expired-request handling
- Permanent preservation of original requests and events

---

## ## 2. Sprint 08 Deliverables

Sprint 08 must produce:

1. Signature-request model
2. Signature-participant model
3. Signer-order enforcement
4. Customer authentication
5. Internal-signer authentication
6. Secure signature links
7. Email-delivery integration
8. Delivery-status tracking
9. Reminder scheduling
10. Seven-day reminder behavior
11. Seventy-five-day expiration behavior
12. Signature capture
13. Signature timestamps
14. Signature audit records
15. Completed-document certificates
16. Signed-PDF generation
17. Signature-request cancellation

18. Signature-request resend
19. Signature-request revision handling
20. Signature-request extension handling
21. Expiration exception tasks
22. Electronic-signature audit events
23. Signature acceptance tests
24. Sprint 09 payment and receivable handoff

---

## ## 3. Requirements Carried Forward

The implementation must preserve these approved requirements:

- Documents are generated from approved Master Sales Record data.
- The document version being signed must remain identifiable.
- The contract is signed by the customer first.
- The contract is signed by GECC second.
- Either the Sales Manager or Comptroller may sign the contract for GECC.
- The GECC signer signs the invoice first.
- The customer signs the invoice second.
- The customer signs the certificate first.
- The Installation Manager signs the certificate second.
- The Comptroller may sign the certificate only when the Installation Manager is unavailable.
- A substitute Comptroller signature requires a recorded reason.
- Signature links are delivered by email.
- Signers must be authenticated.
- Automatic reminders are sent every seven days.
- Signature requests expire after 75 days.
- Expired requests create action tasks for the Sales Manager and Comptroller.
- Original signature requests remain preserved after expiration.
- Authorized users may resend, revise, cancel, or extend requests.
- Reminder and expiration events remain in the audit trail.
- Completed signed PDFs and signature audit certificates are retained.

---

## ## 4. Signature Data Model

### ### 4.1 `signature\_request`

| Field                 | Type        | Requirement                                                                 |
|-----------------------|-------------|-----------------------------------------------------------------------------|
| ---                   | ---         | ---                                                                         |
| `id`                  | BIGINT      | Primary key                                                                 |
| `document_version_id` | BIGINT      | Required foreign key                                                        |
| `project_id`          | BIGINT      | Required foreign key                                                        |
| `msr_version_id`      | BIGINT      | Required foreign key                                                        |
| `request_type`        | VARCHAR(30) | Contract, invoice, certificate                                              |
| `request_status`      | VARCHAR(40) | Draft, pending, partially_signed, completed, expired, cancelled, superseded |
| `created_at`          | TIMESTAMPTZ | Required                                                                    |
| `created_by`          | BIGINT      | Required foreign key                                                        |
| `sent_at`             | TIMESTAMPTZ | Optional                                                                    |

|  |                                  |  |             |  |                      |  |
|--|----------------------------------|--|-------------|--|----------------------|--|
|  | `expires_at`                     |  | TIMESTAMPTZ |  | Required             |  |
|  | `completed_at`                   |  | TIMESTAMPTZ |  | Optional             |  |
|  | `cancelled_at`                   |  | TIMESTAMPTZ |  | Optional             |  |
|  | `cancelled_by`                   |  | BIGINT      |  | Optional foreign key |  |
|  | `cancellation_reason`            |  | TEXT        |  | Optional             |  |
|  | `supersedes_request_id`          |  | BIGINT      |  | Optional foreign key |  |
|  | `extended_from_request_id`       |  | BIGINT      |  | Optional foreign key |  |
|  | `extension_reason`               |  | TEXT        |  | Optional             |  |
|  | `signature_audit_certificate_id` |  | BIGINT      |  | Optional foreign key |  |
|  | `signed_document_version_id`     |  | BIGINT      |  | Optional foreign key |  |
|  | Standard audit columns           |  | —           |  | Required             |  |

### ### 4.2 `signature\_participant`

|  |                         |  |              |  |                                                                               |  |
|--|-------------------------|--|--------------|--|-------------------------------------------------------------------------------|--|
|  | Field                   |  | Type         |  | Requirement                                                                   |  |
|  | ---                     |  | ---          |  | ---                                                                           |  |
|  | `id`                    |  | BIGINT       |  | Primary key                                                                   |  |
|  | `signature_request_id`  |  | BIGINT       |  | Required foreign key                                                          |  |
|  | `signer_role`           |  | VARCHAR(40)  |  | Customer, Sales Manager, Comptroller, Installation Manager                    |  |
|  | `signer_user_id`        |  | BIGINT       |  | Optional foreign key                                                          |  |
|  | `signer_name`           |  | VARCHAR(200) |  | Required                                                                      |  |
|  | `signer_email`          |  | VARCHAR(320) |  | Required                                                                      |  |
|  | `signing_order`         |  | INTEGER      |  | Required                                                                      |  |
|  | `participant_status`    |  | VARCHAR(40)  |  | Pending, invited, viewed, authenticated, signed, declined, expired, cancelled |  |
|  | `required`              |  | BOOLEAN      |  | Required                                                                      |  |
|  | `authentication_method` |  | VARCHAR(50)  |  | Required                                                                      |  |
|  | `invited_at`            |  | TIMESTAMPTZ  |  | Optional                                                                      |  |
|  | `viewed_at`             |  | TIMESTAMPTZ  |  | Optional                                                                      |  |
|  | `authenticated_at`      |  | TIMESTAMPTZ  |  | Optional                                                                      |  |
|  | `signed_at`             |  | TIMESTAMPTZ  |  | Optional                                                                      |  |
|  | `declined_at`           |  | TIMESTAMPTZ  |  | Optional                                                                      |  |
|  | `decline_reason`        |  | TEXT         |  | Optional                                                                      |  |
|  | `ip_reference`          |  | VARCHAR(100) |  | Optional                                                                      |  |
|  | `user_agent_reference`  |  | TEXT         |  | Optional                                                                      |  |

Signer identity and signature evidence must be retained according to the signature-audit requirements.

### ### 4.3 `signature\_field`

|  |                             |  |              |  |                                 |  |
|--|-----------------------------|--|--------------|--|---------------------------------|--|
|  | Field                       |  | Type         |  | Requirement                     |  |
|  | ---                         |  | ---          |  | ---                             |  |
|  | `id`                        |  | BIGINT       |  | Primary key                     |  |
|  | `signature_request_id`      |  | BIGINT       |  | Required foreign key            |  |
|  | `participant_id`            |  | BIGINT       |  | Required foreign key            |  |
|  | `document_field_name`       |  | VARCHAR(150) |  | Required                        |  |
|  | `field_type`                |  | VARCHAR(30)  |  | Signature, date, initials, text |  |
|  | `page_number`               |  | INTEGER      |  | Required                        |  |
|  | `field_coordinates`         |  | JSONB        |  | Required                        |  |
|  | `required`                  |  | BOOLEAN      |  | Required                        |  |
|  | `field_status`              |  | VARCHAR(30)  |  | Pending, completed, voided      |  |
|  | `completed_value_reference` |  | TEXT         |  | Optional                        |  |

| `completed\_at` | TIMESTAMPTZ | Optional |

### ### 4.4 `signature\_link`

| Field             | Type         | Requirement                                |
|-------------------|--------------|--------------------------------------------|
| ---               | ---          | ---                                        |
| `id`              | BIGINT       | Primary key                                |
| `participant_id`  | BIGINT       | Required foreign key                       |
| `link_identifier` | VARCHAR(200) | Required, unique                           |
| `token_hash`      | VARCHAR(128) | Required                                   |
| `created_at`      | TIMESTAMPTZ  | Required                                   |
| `first_used_at`   | TIMESTAMPTZ  | Optional                                   |
| `last_used_at`    | TIMESTAMPTZ  | Optional                                   |
| `expires_at`      | TIMESTAMPTZ  | Required                                   |
| `revoked_at`      | TIMESTAMPTZ  | Optional                                   |
| `link_status`     | VARCHAR(30)  | Active, used, expired, revoked, superseded |

The system must not store reusable signature tokens in plain text.

---

## ## 5. Signing-Order Rules

### ### 5.1 Contract

Required order:

```text

1. Customer signs contract
 2. Sales Manager or Comptroller signs contract
- The GECC signer cannot sign before the customer signature is complete.

5.2 Invoice

Required order:

text

1. Sales Manager or Comptroller signs invoice
 2. Customer signs invoice
- The customer cannot sign before the GECC signature is complete.

The invoice signature request cannot be sent until the contract has all required signatures.

5.3 Certificate

Required order:

text

1. Customer signs certificate
 2. Installation Manager signs certificate
- The Comptroller may substitute only when the Installation Manager is unavailable.

For a substitute Comptroller signature:

The substitute reason is required.

The reason is stored with the signature request.

The reason is stored with the certificate record.

The reason is included in the signature audit certificate.

The substitute action is audited.

5.4 Signer-order enforcement

The system must reject:

Signing before the participant's turn

Accessing an unauthorized signing action

Signing a superseded document version

Signing an expired request

Signing a cancelled request

Signing a request for which authentication has not succeeded

Completing a signature field for another participant

6. Signature Request Creation

A signature request may be created only when:

The document version is current.

The Master Sales Record version is approved.

Required workflow state permits signing.

Required document fields are complete.

Required signer identities are known.

Required signer email addresses are present.

Required signature fields are mapped.

The document content hash is recorded.

The request has not been superseded or cancelled.

The request must preserve:

Document version

Master Sales Record version

Project

Customer

Template version

Artwork version

Document content hash

Participant order

Expiration timestamp

Creation user

Creation timestamp

7. Email Delivery

7.1 Invitation email

Each signer invitation must include:

GECC identification

Document type

Project identification code where appropriate

Signer role

Secure signature link

Expiration information

Support or contact instructions
No unnecessary sensitive information
7.2 Delivery statuses
The system must track:

text

queued
sent
delivered
bounced
blocked
opened
link_accessed
failed

Delivery events must remain attached to the signature request.

7.3 Email records

signature_delivery_event

| Field | Type | Requirement |
|----------------------|--------------|--|
| id | BIGINT | Primary key |
| signature_request_id | BIGINT | Required foreign key |
| participant_id | BIGINT | Required foreign key |
| delivery_type | VARCHAR(40) | Invitation, reminder, expiration, resend |
| recipient_email | VARCHAR(320) | Required |
| sent_at | TIMESTAMPTZ | Optional |
| delivered_at | TIMESTAMPTZ | Optional |
| opened_at | TIMESTAMPTZ | Optional |
| bounced_at | TIMESTAMPTZ | Optional |
| delivery_status | VARCHAR(30) | Required |
| provider_reference | VARCHAR(200) | Optional |
| failure_reason | TEXT | Optional |

Every delivery event must be audited.

8. Signer Authentication

8.1 Required behavior

The system must authenticate signers before allowing signature completion.

Authentication must support:

Secure signature link

Signer identity confirmation

Email confirmation

Additional authentication where configured

Re-authentication for a sensitive signing action where required

Prevention of link reuse after completion or revocation

8.2 Customer authentication

The customer must confirm identity through the configured customer-authentication method before signing.

The exact authentication method remains an implementation configuration item.

Possible methods include:

Email link plus verification code

Email link plus known customer information

One-time code

Approved external identity verification

8.3 Internal signer authentication

Sales Manager, Comptroller, and Installation Manager signers must:

Authenticate through an active GECC user account.

Have the required role and permission.

Sign only when it is their turn.

Sign the exact document version presented.

Have the signature recorded against their user identity.

8.4 Authentication failure

After authentication failure:

Signature is denied.

The event is logged.

The request remains active unless a security rule suspends it.

Repeated failures may trigger a security review or link revocation.

The user is not shown restricted internal information.

9. Signature Capture

The system must capture:

Signer identity

Signer role

Signature image or secure signature representation

Signature timestamp

Signature date

Signature field

Document version

Document content hash

Authentication result

Signature request

Session or signing session reference

Delivery and access evidence

Applicable IP or device reference according to the final implementation

The system must prevent a signer from changing the underlying document after signing.

10. Reminder Process

10.1 Reminder interval

Automatic reminders are sent every seven days while:

The request is active.

The participant has not signed.

The request has not expired.

The request has not been cancelled.

The request has not been superseded.

10.2 Reminder schedule

The first reminder is calculated from the original invitation or prior reminder according to the configured schedule.

The system must store:

- Reminder number
- Scheduled time
- Actual send time
- Recipient
- Delivery result
- Failure reason where applicable

10.3 Reminder limits

The system must stop reminders when:

- The participant signs.
- The request expires.
- The request is cancelled.
- The request is superseded.
- The request is revoked.
- An administrator pauses reminders.

Reminder pausing must require an authorized reason and audit event.

11. Expiration

11.1 Expiration period

Unsigned contracts, invoices, and certificates expire after 75 days.

The expiration timestamp is calculated from the signature-request creation or sending time according to the approved implementation configuration.

The request must store the exact expires_at timestamp.

11.2 Expiration behavior

Upon expiration:

- Mark the request as expired.
- Disable active signing links.
- Stop automatic reminders.
- Preserve the original request.
- Preserve all delivery and access events.
- Create an expiration audit event.
- Create an action task for the Sales Manager and Comptroller.
- Display the request on the exception dashboard.
- Prevent further signing until an authorized action occurs.

11.3 Authorized actions after expiration

An authorized user may:

- Resend the same document version
- Create a new signature request
- Revise the document through the approved workflow
- Cancel the request
- Extend the request
- Record a reason

An expired request must not be silently reactivated without a recorded action.

12. Resend, Revision, Cancellation, and Extension

12.1 Resend

Resending may be used when:

The document version remains valid.

The request is expired, failed, or delivery-blocked.

No source-data revision is required.

A resend must:

Create a new delivery event.

Preserve the original request.

Preserve the original expiration history.

Record the resending user and reason.

Use a new secure link where appropriate.

Not remove prior signer activity.

12.2 Revision

A revision is required when:

Underlying project data changes.

The Master Sales Record version changes.

The document version changes.

Scope changes.

Pricing changes.

Contract, invoice, or certificate terms change.

A revised document requires:

New document version

New signature request

New signing sequence

Preservation of the original request

Preservation of original signatures

Superseding relationship

Audit events

12.3 Cancellation

Cancellation requires:

Authorized user

Cancellation reason

Cancellation timestamp

Link revocation

Reminder cancellation

Audit event

The cancelled request remains preserved.

12.4 Extension

An extension may be granted only by an authorized user.

The extension requires:

Existing request

Extension reason

New expiration timestamp

Authorized user

Approval or authorization

Audit event

The original expiration value remains preserved.

13. Completed Signature Documents

Upon completion of all required signatures:

- Verify signing order.
- Verify all required fields.
- Verify signer identities.
- Verify document content hash.
- Apply signatures to the document.
- Generate the completed signed PDF.
- Generate a signature audit certificate.
- Store both files.
- Link them to the document version.
- Update signature statuses.
- Update the project workflow state.
- Create completion audit events.
- Create the next workflow task where applicable.

13.1 signature_audit_certificate

| Field | Type | Requirement |
|----------------------------|--------------|----------------------|
| id | BIGINT | Primary key |
| signature_request_id | BIGINT | Required foreign key |
| document_version_id | BIGINT | Required foreign key |
| certificate_file_location | TEXT | Required |
| certificate_content_hash | VARCHAR(128) | Required |
| generated_at | TIMESTAMPTZ | Required |
| signer_summary | JSONB | Required |
| delivery_summary | JSONB | Required |
| authentication_summary | JSONB | Required |
| signature_sequence_summary | JSONB | Required |
| completion_status | VARCHAR(30) | Required |

13.2 Certificate content

The certificate should include:

- Document identity
- Document version
- Project identification code
- Master Sales Record version
- Signer names
- Signer roles
- Signature order
- Signature timestamps
- Authentication method
- Delivery evidence
- Reminder history
- Document content hash
- Completion timestamp
- Substitute-signature reason where applicable

The final certificate layout remains an implementation configuration item.

14. Signature Statuses

- Request statuses
- text

draft
pending
partially_signed
completed
expired
cancelled
superseded
failed
Participant statuses
text

pending
invited
delivered
viewed
authenticated
signed
declined
expired
cancelled
Field statuses
text

pending
completed
voided
Status changes must be controlled and audited.

15. Workflow Integration

Contract

When the customer signs:

Update contract participant status.
Permit GECC signing.
Create the GECC signature task.
Audit the event.

When the GECC signer signs:

Mark the contract fully signed.
Update the project workflow.
Permit invoice-signature workflow.

Invoice

When the GECC signer signs:

Permit customer invoice signing.
When the customer signs:

Mark the invoice fully signed.
Calculate the installation waiting-period start timestamp.

Permit installation-task creation when all other requirements are satisfied.

Certificate

When the customer signs:

Permit Installation Manager signing.

When the Installation Manager signs:

Mark certificate fully signed.

When the Comptroller substitutes:

Confirm Installation Manager unavailability.

Require substitute reason.

Store substitute reason.

Mark the certificate fully signed after valid signature.

Audit the substitution.

16. Access and Permission Rules

Customer

May:

Access only their own signature request.

View the document presented for signature.

Sign only when it is their turn.

Download the completed document if permitted.

Receive reminders and expiration notices.

May not:

View internal approval notes.

View internal commission or financial information.

Alter document source data.

Sign on behalf of another participant.

Sales Manager and Comptroller

May:

Sign contracts.

Sign invoices.

Approve or manage expired requests.

Resend, cancel, revise, or extend requests where authorized.

View signature status for authorized business records.

Approve certificate substitution according to applicable rules.

Installation Manager

May:

Sign certificates.

View authorized certificate status.

Manage installation-related signature tasks.

Sales Associate

May:

View signature status for assigned projects.

Review generated documents where authorized.

Send permitted signature requests.

Not sign for GECC unless separately authorized.

Technician

May:

Review and send certificates to customers where authorized.

View certificate status for assigned projects.

Not sign as GECC.

Not alter document content.

17. Audit Events

Sprint 08 must create audit events for:

Signature request created

Participant invited

Email queued

Email sent

Email delivered

Email bounced

Email opened

Signature link accessed

Authentication succeeded

Authentication failed

Document viewed

Signature field completed

Signature completed

Signature declined

Reminder scheduled

Reminder sent

Reminder failed

Signature request expired

Signature request cancelled

Signature request resent

Signature request revised

Signature request extended

Signature request superseded

Signed PDF generated

Signature audit certificate generated

Signature request downloaded

Unauthorized signature attempt

Out-of-order signing attempt

Expired-link attempt

Revoked-link attempt

Substitute certificate signature

Signature completion failure

Each event must reference:

User or external signer identity where available

Signer role

Session or signing session

Signature request

Participant

Project

Customer

Document version

Master Sales Record version

Event outcome

Failure reason where applicable

Timestamp

Audit-chain references

18. Sprint 08 Decisions

D-08-001 — Signature-request version binding

Every signature request is bound to exactly one immutable document version and one Master Sales Record version.

D-08-002 — Signing order

Signing order is enforced by document type:

Contract: customer first, GECC second

Invoice: GECC first, customer second

Certificate: customer first, Installation Manager second

D-08-003 — Comptroller substitute signing

The Comptroller may sign a certificate only when the Installation Manager is unavailable and a substitute reason is recorded.

D-08-004 — Reminder interval

Automatic reminders use a seven-day interval.

D-08-005 — Expiration period

Unsigned requests expire after 75 days.

D-08-006 — Expiration preservation

Expired requests remain permanently preserved. Expiration does not delete or overwrite the original request.

D-08-007 — Secure links

Signature links use revocable, expiring, non-reusable token references. Plain-text reusable tokens are not stored.

D-08-008 — Internal signer authentication

Internal GECC signers authenticate through active system accounts with the required role and permission.

D-08-009 — Customer authentication

Customers must authenticate through the approved customer-authentication method before signing.

D-08-010 — Revised documents

A revised document requires a new signature request and a new signing sequence where applicable.

D-08-011 — Signed-PDF preservation

Completed signed PDFs and signature audit certificates are permanently retained and linked to the original document version.

D-08-012 — Signature audit integration

All signature, delivery, authentication, reminder, expiration, and failure events are written to the tamper-evident audit log.

19. Sprint 08 Acceptance Criteria

Sprint 08 is accepted when:

- A signature request can be created for a valid contract.
- A signature request can be created for a valid invoice.
- A signature request can be created for a valid certificate.
- Each request is bound to one document version.
- Each request is bound to one Master Sales Record version.
- Contract signing requires customer signature before GECC signature.
- Invoice signing requires GECC signature before customer signature.
- Certificate signing requires customer signature before GECC signature.
- Installation Manager certificate signing is supported.
- Comptroller certificate substitution requires Installation Manager unavailability and a reason.
- Out-of-order signing is rejected.
- Unauthorized users cannot sign.
- Signers are authenticated before signing.
- Secure signature links are generated.
- Signature links expire or are revoked as required.
- Email delivery statuses are tracked.
- Automatic reminders are sent at seven-day intervals.
- Reminders stop after signing, cancellation, supersession, or expiration.
- Requests expire after 75 days.
- Expired requests cannot be signed.
- Expired requests create Sales Manager and Comptroller action tasks.
- Expired requests remain preserved.
- Authorized users can resend requests.
- Authorized users can cancel requests.
- Authorized users can extend requests with a reason.
- Revised documents create new signature requests.
- Original signature requests remain preserved.
- All required signature fields are completed before completion.
- Signed PDFs are generated and stored.
- Signature audit certificates are generated and stored.
- Signed PDFs include the correct document version.
- Signature certificates include signer and delivery evidence.
- Signature completion updates the project workflow.
- All signature events are audited.
- All delivery events are audited.
- All authentication failures are audited.
- Unauthorized and out-of-order attempts are audited.

Sprint 09 can rely on completed invoice-signature state before processing payments.

20. Sprint 08 Test Cases

Request creation

- Create a contract signature request from a valid document.
- Create an invoice signature request only after the contract is fully signed.
- Create a certificate signature request from a valid certificate.
- Reject a request for an unapproved document.
- Reject a request for a superseded document version.
- Reject a request with missing signer data.
- Reject a request with missing signature fields.

Signing order

- Customer signs contract.
- GECC signs contract after customer.
- Reject GECC contract signing before customer.

GECC signs invoice.
Customer signs invoice after GECC.
Reject customer invoice signing before GECC.
Customer signs certificate.
Installation Manager signs certificate after customer.
Reject Installation Manager signing before customer.
Comptroller substitute signature requires a reason.
Reject unauthorized substitute signature.

Authentication
Valid internal signer authenticates.
Invalid internal signer authentication is rejected.
Valid customer authentication succeeds.
Invalid customer authentication is rejected.
Expired link is rejected.
Revoked link is rejected.
Used completed link cannot be reused.

Reminders and expiration
Invitation is sent.
Delivery failure is recorded.
Seven-day reminder is scheduled.
Reminder is sent.
Reminder stops after signing.
Request expires after 75 days.
Expired request creates action tasks.
Expired request remains preserved.
Authorized extension updates expiration and preserves original values.

Completed documents
All signature fields are completed.
Signed PDF is generated.
Signature certificate is generated.
Signed PDF content hash is recorded.
Certificate includes signer order.
Certificate includes signature timestamps.
Certificate includes delivery and authentication evidence.
Completed request updates project workflow.

Revision and preservation
Revise a signed source document.
Preserve original signed PDF.
Preserve original signature certificate.
Create a new document version.
Create a new signature request.
Require a new signing sequence.
Mark the original request superseded.
Preserve all original audit events.

Audit and security
Signature creation is audited.
Signature access is audited.
Authentication failure is audited.
Out-of-order signing is audited.
Expiration is audited.
Reminder delivery is audited.
Resend is audited.
Cancellation is audited.

Unauthorized download is rejected and audited.

21. Sprint 08 Open Questions

These implementation questions do not change approved requirements:

What exact customer-authentication method will be used?

What signature representation will be accepted?

What identity-verification level is required for customers?

What email delivery service will be used?

What exact expiration start point will be used: request creation, first delivery, or successful delivery?

What maximum number of reminders is permitted?

What exact reminder time will be used in Eastern time?

What signer notification will be sent after a failed delivery?

What constitutes Installation Manager unavailability?

Who may approve an extension beyond 75 days?

What signature-audit-certificate format is required?

What legal wording must appear on completed signature certificates?

What retention and backup requirements apply to signature evidence?

What exact customer support process handles disputed signatures?

What additional authentication is required for certificate substitute signing?

22. Sprint 09 Handoff

Sprint 09 will implement:

Cashier's-check receipt

Bank-wire receipt

Payment verification

Payment allocation

Accounts receivable

Credits

Refunds

Adjustments

Separate accounts payable records

Sprint 09 must use:

Fully signed invoice status

Invoice document version

Project identification code

Customer and project references

Approved adjustment workflow

Signature audit certificate

Payment-related task structure

Sprint 09 must not treat an invoice as payable or a project as complete unless:

Required invoice signatures are complete.

Payment is recorded.

Payment verification is complete.

Approved adjustments are properly recorded.

23. Historical Record

Supersedes

None.

Preserves

GECC-00-project-baseline.md

GECC-01-data-model.md
GECC-02-auth-and-permissions.md
GECC-03-audit-log.md
GECC-04-customers-and-projects.md
GECC-05-master-sales-record.md
GECC-06-workflow-state-machine.md
GECC-07-document-generation.md
Sprint 00 decisions
Sprint 01 decisions
Sprint 02 decisions
Sprint 03 decisions
Sprint 04 decisions
Sprint 05 decisions
Sprint 06 decisions
Sprint 07 decisions
Approved signing order
Approved reminder and expiration rules
Approved document-version and audit requirements
Changes introduced by this bundle
Defines the electronic-signature request model.
Defines participant, field, link, delivery, and audit-certificate structures.
Enforces signing order for contracts, invoices, and certificates.
Defines signer authentication behavior.
Defines seven-day reminders.
Defines 75-day expiration.
Defines resend, revision, cancellation, and extension behavior.
Defines completed signed PDFs and signature audit certificates.
Defines Sprint 08 audit events.
Defines Sprint 09 payment and receivable handoff.
===== END FILE: GECC-08-Electronic-Signatures.md =====
=== BEGIN FILE: GECC-09-Payments-And-Payables.md ===
Bundle name: GECC-09-payments-and-payables.md
Project: GECC Sales Back Office System
Company: Go Ecco Climate Control
Sprint: 09
Sprint name: Payments, Receivables, Payables, and Adjustments
Baseline: GECC-00-project-baseline.md
Status: Sprint 09 implementation bundle
Date: 2026-09-20
Previous bundle: GECC-08-electronic-signatures.md
Next bundle: GECC-10-costs-and-commissions.md
Primary database: Relational database
System of record: GECC database
Google Sheets: Export or reporting destination only
1. Sprint Objective
Implement the payment, accounts-receivable, accounts-payable, refund, credit, and adjustment functions required to support the approved GECC project lifecycle.

Sprint 09 must ensure that:

Payments are allocated to the correct customer project and invoice.
Unverified payments cannot satisfy payment requirements.
Cashier's checks require bank confirmation.

Bank wires require a wire reference number or proof of receipt.

Unpaid invoices remain visible as accounts receivable.

Separate payables are created for equipment, dealer fees, and materials.

Payables are created when the Certificate is fully signed, regardless of when vendors are paid.

Credits, refunds, and adjustments require proper approval.

All financial actions are permission-controlled and auditable.

Payment status correctly controls project completion.

2. Sprint Scope

2.1 Included

Payment records

Cashier's-check workflow

Bank-wire workflow

Payment verification

Payment allocation

Partial payments

Overpayment detection

Accounts-receivable balances

Credits

Refunds

Invoice adjustments

Automatic payable creation

Equipment payables

Dealer-fee payables

Materials payables

Payable approval

Payable payment recording

Payment and payable statuses

Cash-basis accounting events

Financial exception handling

Receivable and payable reports

Audit-log integration

Dashboard integration

2.2 Excluded

Automatic bank-statement import

Automatic bank reconciliation

Automatic bank-transaction matching

Credit-card processing

ACH processing

Customer payment portal

Vendor portal

Payroll processing

Commission calculations

Final financial-statement implementation

3. Sprint Deliverables

3.1 Payment Data Model

Implement database structures for:

Payment

Payment verification

Payment allocation

Payment reversal

Payment refund

Payment evidence or proof-of-receipt document

Required payment fields include:

Payment ID
Customer ID
Project ID
Invoice ID
Project identification code
Payment method
Amount
Payment date
Received date
Verification status
Payment status
Cashier's-check number
Bank name
Bank-wire reference number
Proof-of-receipt document reference
User entering payment
User verifying payment
Verification date and time
Notes
Related adjustment reference
Created date and time
Last modified date and time

3.2 Payment Methods

The initial implementation must support:

Bank cashier's check

Bank wire transfer

Unsupported methods must be rejected by validation and database constraints.

3.3 Payment Verification

Implement verification workflows for:

Cashier's checks

Bank wires

A payment must remain unavailable for project completion and accounts-receivable reduction until it is verified.

Cashier's-check verification requires:

Cashier's-check number

Bank name

Bank confirmation

Verification user

Verification date and time

Bank-wire verification requires at least one of:

Manually entered wire reference number

Uploaded proof of receipt

3.4 Payment Allocation

Each payment must be allocated to: